

technologies such as 2.5G, 2.75G, 3G and 4G; however, 2G networks are still used in many parts of the world.

3G, short for third generation, is the third generation of mobile telecommunications technology. The first 3G networks were introduced in 1998. This technology is based on a set of standards used for mobile devices, and mobile telecommunications use services and networks that comply with International Mobile Telecommunications-2000 (IMT-2000) standard specifications by International Telecommunication Union. 3G finds application in wireless voice telephony, mobile Internet access, fixed wireless Internet access, video calls and mobile TV.

3G telecommunication networks support services that provide an information transfer at the rate of at least 200kbps. Later 3G releases, often denoted as 3.5G and 3.75G, provide mobile broadband access of several Mbps to smartphones and mobile modems in laptop computers. This ensures it can be applied to wireless voice telephony, mobile Internet access, fixed wireless Internet access, video calls and mobile TV technologies.

Fourth generation, or 4G, the next generation of mobile telecommunications technology, succeeding 3G and preceding 5G, was introduced in 2008. A 4G system, in addition to the usual voice and other services of 3G, provides mobile broadband Internet access, for example, to laptops with wireless modems, smartphones and other mobile devices.

Potential and current applications include amended mobile Web access, IP telephony, gaming services, high-definition mobile TV, video conferencing, 3D television and cloud computing. Two 4G candidate systems that have been commercially deployed are mobile WiMAX standard and the first-release Long Term Evolution (LTE) standard.

Expected 5G network visions

Rapid development of wireless technologies coupled with standards convergence herald the emergence of fifth-generation (5G) wireless communication. Broadly speaking, 5G is expected to provide much greater capacity to meet growing user demand resulting from a number of new services compared to 4G.

5G (fifth-generation mobile networks or fifth-generation wireless systems) denotes the next major phase of mobile telecommunications standards beyond the current 4G/IMT-Advanced standards. Experts feel that 5G should be rolled out by 2020 to meet business and consumer demands.

The major difference from a user point of view between 4G and 5G techniques must be something else than increased peak bit rate. It could be higher number of simultaneously-connected devices, higher system-spectral efficiency (data volume per area unit), lower battery consumption, lower outage probability (better coverage), high bit rates in larger portions of coverage area, lower latencies, higher number of supported devices, lower infrastructure deployment costs, higher versatility and scalability or higher reliability of communications.

In addition to simply providing faster speeds, experts predict that 5G networks will also need to meet the needs of new use-cases such as the Internet of Things (IoT) as well as broadcast-like services and lifeline communications in times of natural disasters. The Next Generation Mobile Networks Alliance defines the following requirements for 5G networks:

1. Data rates of several tens of Mbps should be supported for tens of thousands of users
2. 1Gbps to be offered simultaneously to tens of workers on the same office floor
3. Several hundreds of thousands of simultaneous connections to

be supported for massive sensor deployments

4. Spectral efficiency should be significantly enhanced compared to 4G
5. Coverage should be improved
6. Signalling efficiency should be enhanced
7. Latency should be significantly reduced compared to LTE

Expectations are that 5G will provide uniform throughput of at least 1Gbps, peaking at around 10Gbps, with a couple of milliseconds of latency, offering a highly-reliable service. 5G will provide a truly ubiquitous unlimited mobile experience through terminals enhanced with artificial intelligence (AI) capabilities.

New applications are foreseen that will facilitate domains such as e-health and machine-to-machine (M2M) communication. Its salient features would include:

1. A super-efficient mobile network that delivers a better performing network at lower investment costs. It would address mobile network operators' pressing need to see the unit cost of data transport falling at roughly the same rate as the volume of data demand is rising. It would be a leap forward in efficiency based on IET demand attentive network (DAN) philosophy.
2. A super-fast mobile network comprising the next generation of small cells densely clustered together to give a contiguous coverage over at least urban areas and gets the world to the final frontier for true wide area mobility. It would require access to spectrum under 4GHz, perhaps via the world's first global implementation of dynamic spectrum access.
3. A converged fibre wireless network that uses, for the first time for wireless Internet access, millimetre wave bands (20GHz-60GHz) so as to allow very wide